DSP Lab 2 Delay and Echo

I. Objectives

1. Get familiar with Matlab and DSK6713.
2. Learn basic programming on DSK6713.
3. Learn to process acoustic signals with DSK6713.

II. Procedures

A. Get familiar with acoustic signal processing with DSK6713.

1. Open the attached template project tone.pjt with CCStudio 3.1.
2. Connect the port "line in" (J1) of the DSK board to a music source (e.g. mp3 player, or the sound output of your computer) by using the male-male 3.5mm stereo cable.
3. Connect the port "head phone" to a speaker or headphone.
4. Load the program to the DSK board, and execute it.
5. Play music at the music source, and you should be able to hear it through the speaker connected to the DSK board. The template program pass the music directly through without any processing.
6. Read the source file "main.c", and understand the operations.
7. Read the source file "bargraph.c". This function can indicate the sound volume by using the four LEDs on the DSK (more LEDs lit means larger volume).

B. Delay and Echo

We are going to simulate delays in communications networks and reflection of sound heard in a canyon.

8. Write a function "delay". The inputs of the function are: one sample of the sound (signed int, from either left channel or right channel), the amount of time to be delayed (unsigned int, in the unit ms). The output of the function is a delayed sound sample (signed int) according to the delay parameter at the function input. The maximum delay is 1 second.
    a. Hint: You can use a global circular buffer and a static index to achieve this function.
    b. Hint: store the input sample at the top of the circular buffer (the static index variable can be used to point to the top of the buffer).
    c. Hint: Get the delayed sample from the circular buffer based on the delay parameter.
9. In main.c, add the delayed output to the current input to achieve the effects of echo. E.g. left_output = left_input + delay(left_input, delay_time).
10. Play the music at a large volume. In addition to the music, you will hear some clicks occasionally. Reduce the volume of the music and listen again. Explain the cause of the clicks. Modify your program to fix the problem.
11. Change the value of the delay_time, and you can hear different echo effects.